

DOMINIK HÜPPIN

MSc ETH in Physics

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Physicist with experience in numerical modelling, electromagnetic simulation, data analysis, Python-based automation, and experimental research. International research experience at ETH Zürich, Kyoto University, and ISAS/JAXA.

EDUCATION

- 07/2025–09/2026 **ETH Zürich** · MSc Physics GPA 5.44/6.0  Zürich, Switzerland
Focus: Astrophysics and Quantum Field Theory
- 09/2020–07/2025 **ETH Zürich** · BSc Physics GPA 5.0/6.0  Zürich, Switzerland
Focus: Quantum Mechanics and Astrophysics

RESEARCH & EXPERIMENTAL EXPERIENCE

- 04/2026–09/2026 **Sirin Orbital Systems AG / ETH Zürich / Kyoto University** · Master's Thesis (Grade: 6.0/6.0, nominated for the ETH Medal)  Zürich, Switzerland & Kyoto, Japan
- Designed, simulated, fabricated, and experimentally characterized a 61 GHz mechanically beam-steerable travelling-wave slot-array antenna for wireless power transfer.
- Combined antenna design and electromagnetic simulation with prototype fabrication and experimental validation.
- 10/2025 **Institute of Space and Astronautical Science (ISAS/JAXA)** · Skills Acquisition Programme  Sagami-hara, Japan
- Worked in Prof. Koji Tanaka's laboratory on microwave wireless power transfer for space applications.
- Designed and simulated a 5.8 GHz antenna array for space applications.
- 09/2025 **Kyoto University, Research Institute for Sustainable Humanosphere (RISH)** · Semester Project  Kyoto, Japan
- Designed, simulated, fabricated, and experimentally characterized a microwave RF-to-DC rectifier for wireless power transfer in Prof. Naoki Shinohara's laboratory.
- Optimized RF-to-DC conversion efficiency through iterative simulation, fabrication, and measurement.
- 02/2024–06/2024 **ETH Zürich, Hybrid Quantum Systems Group** · Research Project  Zürich, Switzerland
- Developed Python-based automation for data acquisition in single-photon transduction experiments.
- Implemented automated measurements and histogram generation, improving measurement efficiency and reproducibility.

SKILLS & LANGUAGES

Programming: Python, C++, MATLAB, L^AT_EX
Simulation: CST Studio Suite, Advanced Design System (ADS)
Software: Microsoft Office
Languages: German (native), English (C1), Spanish (B2), French (basic)

PERSONAL INTERESTS

Photography, Football, Endurance Sports, Surfing